

Dhruv Venkat

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Education

University of Waterloo

Bachelor of Applied Science, Computer Engineering (Honours)

Sep 2024 - 2029

Waterloo, ON

Professional Experience

Red Lane Group

Software Development Intern

May 2026 - Present

Calgary, AB

- Building a **Claude-backed orchestration pipeline** for support triaging, projecting **100+ hours saved/month**
- Modernizing legacy call transcription with a real-time **compliance recording bot for 100% of support calls**
- **Reduced per-call post-processing latency by 20+ minutes** using real-time streaming SDK transcription

Scotiabank

Software Engineering Intern

Sep 2025 - Dec 2025

Toronto, ON

- Built **LangChain and FAISS RAG agents** to generate technical requirements from customer interactions
- Reduced requirements-gathering time by 40% and **increased developer productivity by 25%**
- Architected a Tesseract OCR pipeline that **increased institutional document throughput by 3x**
- Upgraded 8 enterprise applications from Java 11 to Java 21 for LTS, **reducing future maintenance by 50%**

Tangerine

DevOps Engineering Intern

Jan 2025 - Apr 2025

Toronto, ON

- Built a **GitHub Copilot usage tracker using REST APIs** and Apache Groovy, **saving \$1000+ monthly**
- Deployed Python vulnerability scans across **100+ GitHub repositories**, reducing severe issues by 20%
- Maintained **multi-branch Jenkins CI/CD pipelines** for development teams, **increasing reliability by 30%**

Projects

Firmware Developer (UW Orbital) | Embedded C, Python, FreeRTOS, Git [URL](#)

- Designed and optimized CubeSAT real-time control software as part of the embedded firmware subteam
- **Authored and maintained extensive documentation** for interfaces between hardware and ground control
- Wrote drivers for the LM75BD sensor to handle **real-time temperature monitoring and cooling response**

flowstate: Terminal-Native C++ Code Editor | C++, POSIX TTY, CMake, Git [URL](#)

- Built a **terminal-native C++ code editor** with raw-mode input handling and custom rendering
- Built **syntax highlighting and LSP-backed diagnostics** for C/C++, Python, TypeScript, Go, and Rust
- Designed a **Codex-powered AI assistant** for contextual explanations, code generation, and in-place edits

linkdrop: LAN-Based Link Sharing Daemon | Go, Linux, BASH, SQLite, HTTP [URL](#)

- Built a Go daemon for **sending links from mobile devices to Linux PCs** over LAN without using cloud sync
- Implemented **HTTP endpoints and a lightweight browser UI** to receive, save, and open links with one action
- Designed a schema for a persistent SQLite database to store shared links and relevant metadata
- Wrote **custom Linux installation scripts to provision local application data** and register the daemon as a systemd service

Technical Proficiencies

Languages: C++, Embedded C, Python, TypeScript, Java, Go, SQL, RISC-V Assembly

Libraries and Frameworks: CMake, Node.js, ReactJS, Next.js, Scikit-Learn, TensorFlow, Pandas, FastAPI

Cloud/DevOps: AWS, Microsoft Azure, Git, Jenkins, PostgreSQL, GraphQL